Lambda 1 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 17 complete ICs for all stages within ANN10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
1 ANN10	**base10k**	0.0312	0.126	1.14	1.02	0.164	1	0.0957	1.02	1.19	0	18/18
1 ANN10	**soft10k**	0.204	0.799	4.12	0.99	0.613	2.46	0.265	2.84	2.51	0	17/18
1 ANN10	**long**	0.172	0.683	3.8	0.963	0.524	1.81	0.203	6.25	27.4	0	18/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 64, 68, 69, 73, 74, 76, 78, 88. Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda 1 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 18 complete ICs for all stages within P-only DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
2 P-only DeepONet10	**base10k**	0.0578	0.114	0.951	0.994	0.165	0.993	0.123	0.995	1.19	0	18/18
2 P-only DeepONet10	**soft10k**	0.0793	0.461	0.908	5.52	0.575	0.97	0.102	6.43	2.05	0	18/18
2 P-only DeepONet10	**long**	0.145	0.168	1.36	0.746	0.523	0.554	0.087	1.25	1.05	0	18/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88. Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda 1 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 18 complete ICs for all stages within ANN10 + P-only DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
3 ANN10 + P-only DeepONet10	**base10k**	0.0373	0.0777	0.901	0.92	0.244	0.993	0.0941	0.995	1	0	18/18
3 ANN10 + P-only DeepONet10	**soft10k**	0.0437	0.454	0.69	5.28	0.594	0.985	0.0777	2.95	1.44	0	18/18
3 ANN10 + P-only DeepONet10	**long**	0.0707	0.176	0.659	0.775	0.436	0.735	0.0772	1.13	1.03	0	18/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88. Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda 1 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 8 complete ICs for all stages within T/P DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
4 T/P DeepONet10	**base10k**	0.13	0.647	1.63	0.81	0.543	1	0.244	1.1	0.999	0	17/18
4 T/P DeepONet10	**soft10k**	0.293	0.749	3.78	1.09	0.544	1.11	0.496	7.02	22.7	0	9/18
4 T/P DeepONet10	**long**	0.34	0.946	1.85	1.48	0.578	1.04	0.676	47.9	25.7	0	14/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 39, 44, 45, 64, 68, 69. Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda 1 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 18 complete ICs for all stages within ANN10 + T/P DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
5 ANN10 + T/P DeepONet10	**base10k**	0.0491	0.0957	0.962	0.82	0.174	0.97	0.066	0.921	0.993	0	18/18
5 ANN10 + T/P DeepONet10	**soft10k**	0.0803	0.41	0.932	0.819	0.535	0.822	0.103	0.938	1.16	0	18/18
5 ANN10 + T/P DeepONet10	**long**	0.0902	0.189	0.888	0.588	0.4	0.518	0.0959	0.905	0.992	0	18/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 57, 64, 68, 69, 73, 74, 76, 78, 88. Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda 1 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 8 complete ICs for all stages within T/P/ϕ DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
6 T/P/ϕ DeepONet10	**base10k**	0.11	0.123	1	1	0.244	1	1	1.2	3.79	0	17/18
6 T/P/ϕ DeepONet10	**soft10k**	0.697	0.966	21.1	70.9	0.661	7.13	0.72	8.02	1	0	11/18
6 T/P/ϕ DeepONet10	**long**	0.201	0.956	11.7	2.53	0.458	3.07	0.646	35.6	5.64	0	13/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 39, 53, 64, 69, 78. Team-specific cohorts differ; do not rank architectures using different IC sets.

Lambda 1 continuation | Cantera vs NN NRMSE

Median of per-IC NRMSE on the same 17 complete ICs for all stages within ANN10 + T/P/ϕ DeepONet10

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Stage	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	Complete ICs
7 ANN10 + T/P/ϕ DeepONet10	**base10k**	0.0672	0.121	1	0.606	0.21	1	0.245	0.998	1	0	18/18
7 ANN10 + T/P/ϕ DeepONet10	**soft10k**	0.395	0.881	2.89	3.51	0.521	8.19	0.507	1.22	1.15	0	17/18
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.209	0.347	1.2	1.07	0.549	1.15	0.232	1.29	1.27	0	18/18

Exact user util.py RMS normalization: equal time-sample weights; raw dimensionless ratio without x100.
—: incomplete, no full-horizon score. N/A: empty common cohort or undefined reference RMS. Constant nonzero N2 is valid.
Short10000 updates; long endpoints have different update counts. Both models in every mixed team use the same stage/lambda.
Common ICs: 4, 5, 7, 18, 32, 39, 44, 45, 53, 64, 68, 69, 73, 74, 76, 78, 88. Team-specific cohorts differ; do not rank architectures using different IC sets.